

28-OCT-2025

Project Plan

Task	Description	October				November				December		
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3
1. Project Familiarization	Review current codebase, structure, data preprocessing, training pipeline	X	X	X	X							
2. Finetune ResNet50 (ISIC 2024 / Melanoma Kaggle)	Train and validate baseline deterministic ResNet50				X	X						
3. Variational Layers into ResNet	Add stochastic/variational layers (visible & hidden layers)					X	X	X	X			
4. Compare to Existing Methods (Grad-CAM)	Evaluate uncertainty visualization and compare with Grad-CAM methods						X	X	X	X		
5. Extend Uncertainty to Other Architectures	Apply same uncertainty method to EfficientNet, DenseNet, InceptionNet									X	X	X
6. Final Evaluation & Reporting	Aggregate results, prepare visualizations and documentation										X	X

28-OCT-2025

Summary:

- Discuss the workflow of code with Tyson (NRC) for the different DNNs, training set up, finetuning and tests
- Running ResNet50 finetuning with ISIC_2024 / Kaggle_Melanoma

Blockers:

- Issue with gcloud VM ssh through vscode
 - Set up with information from Pasindu, was able to connect through VM on the web but not through vscode (getting an error for publickey on gcloud) – contacted Pasindu for troubleshooting.

04-NOV-2025

Project Plan

Task	Description	October				November				December		
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3
1. Project Familiarization	Review current codebase, structure, data preprocessing, training pipeline	X	X	X	X							
2. Finetune ResNet50 (ISIC 2024 / Melanoma Kaggle)	Train and validate baseline deterministic ResNet50				X	X						
3. Variational Layers into ResNet	Add stochastic/variational layers (visible & hidden layers)					X	X	X	X			
4. Compare to Existing Methods (Grad-CAM)	Evaluate uncertainty visualization and compare with Grad-CAM methods						X	X	X	X		
5. Extend Uncertainty to Other Architectures	Apply same uncertainty method to EfficientNet, DenseNet, InceptionNet									X	X	X
6. Final Evaluation & Reporting	Aggregate results, prepare visualizations and documentation										X	X

04-NOV-2025

Summary:

- Issues resolved regarding gcloud VM SSH through vscode
- Ran baseline ResNet50 finetuning with ISIC_2024 / Kaggle_Melanoma
- Ran ResNet50 + Variational Last layer for uncertainty estimation
- Review data from ISIC database (clinical non-dermascope photos)

Blockers:

- None

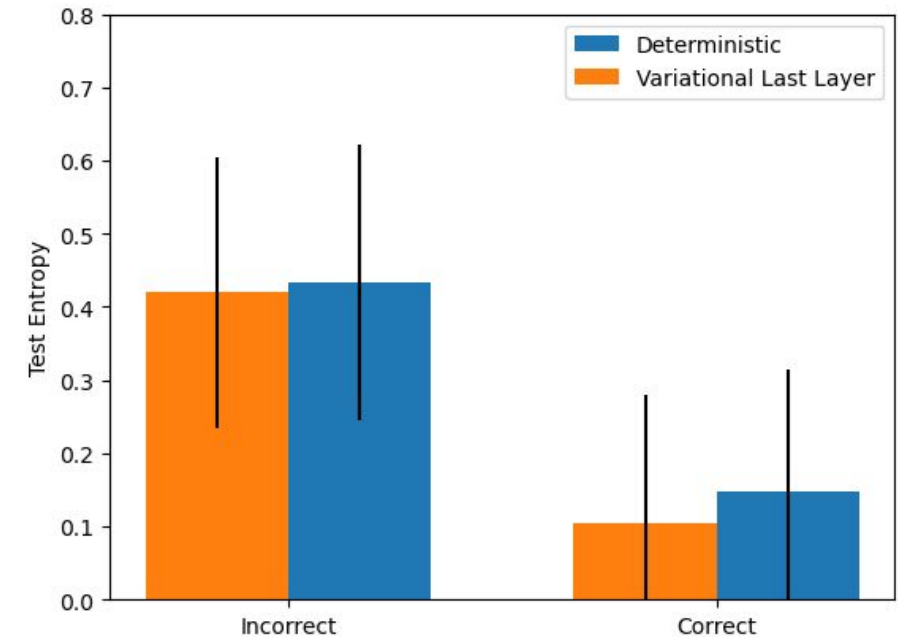
Next Steps:

- Variational visible layers (IP)
- Define an OOD dataset

04-NOV-2025

Variational Last Layer

- Fine tune ResNet50 with ISIC-2024 + Melanoma datasets
- Reparameterize last layer as variational layer



Model	Accuracy
ResNet50 + MLP	0.893
ResNet50 + MLP + VLL	0.913

11-NOV-2025

Project Plan

Task	Description	October				November				December		
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3
1. Project Familiarization	Review current codebase, structure, data preprocessing, training pipeline	X	X	X	X							
2. Finetune ResNet50 (ISIC 2024 / Melanoma Kaggle)	Train and validate baseline deterministic ResNet50				X	X						
3. Variational Layers into ResNet	Add stochastic/variational layers (visible & hidden layers)					X	X	X	X			
4. Compare to Existing Methods (Grad-CAM)	Evaluate uncertainty visualization and compare with Grad-CAM methods						X	X	X	X		
5. Extend Uncertainty to Other Architectures	Apply same uncertainty method to EfficientNet, DenseNet, InceptionNet									X	X	X
6. Final Evaluation & Reporting	Aggregate results, prepare visualizations and documentation										X	X

11-NOV-2025

Summary:

- Ran ResNet50 + Variational Visible layers for uncertainty estimation
- Review data from ISIC database (clinical non-dermascope photos)
 - Create a subset for training the ResNet50 –discuss

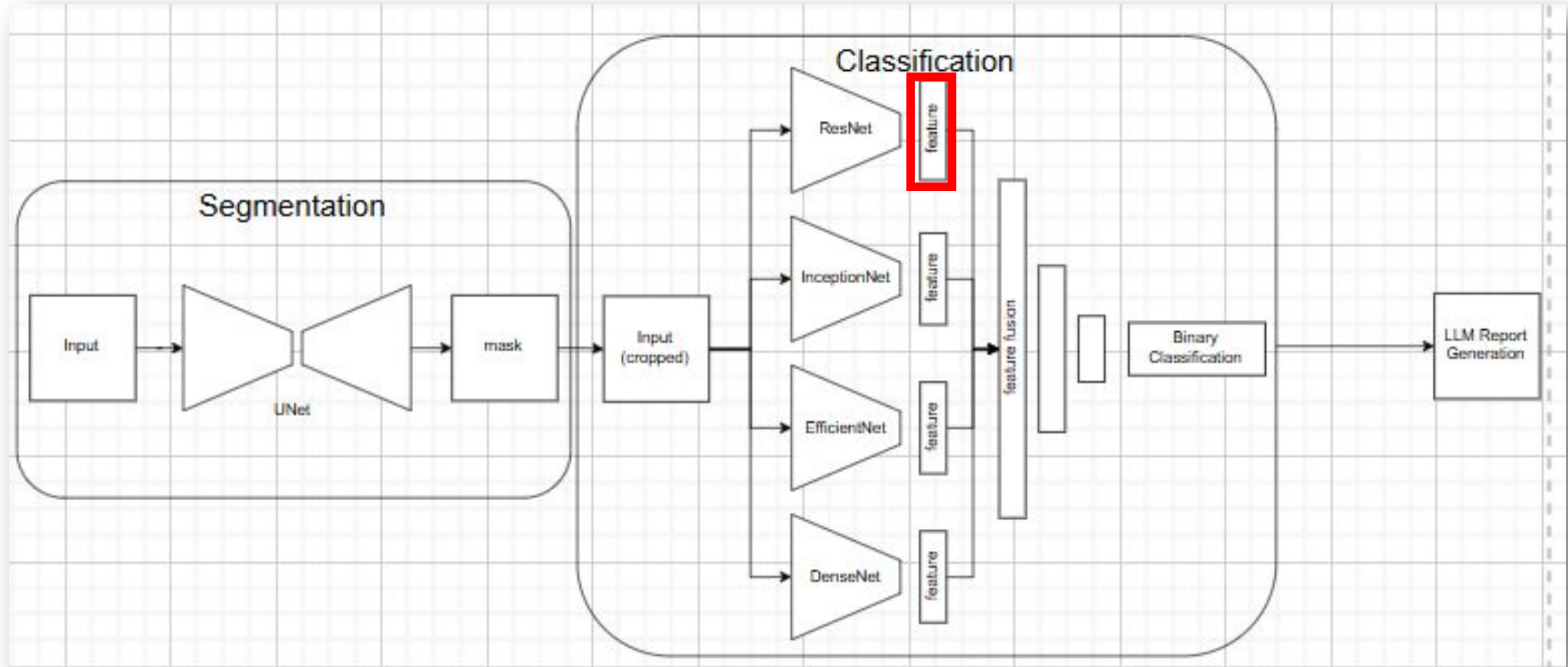
Blockers:

- None

Next Steps:

- Variational complete MLP
- Define an OOD (test) dataset

11-NOV-2025



ISIC - Database

- Clinical
 - Overview: 432
 - 284 malignant
 - Closeup: 9316
 - 4787 malignant
- Useful for customizing our dataset for our use case



18-NOV-2025

Project Plan

		October				November				December		
Task	Description	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3
1. Project Familiarization	Review current codebase, structure, data preprocessing, training pipeline	X	X	X	X							
2.A. Finetune ResNet50 (ISIC 2024 / Melanoma Kaggle)	Train and validate baseline deterministic ResNet50				X	X						
2.B. Finetune ResNet50 on ISIC Clinical close ups								X				
3. Variational Layers into ResNet	Add stochastic/variational layers (visible & hidden layers)					X	X	X	X			
4. Compare to Existing Methods (Grad-CAM)	Evaluate uncertainty visualization and compare with Grad-CAM methods								X	X		
5. Extend Uncertainty to Other Architectures	Apply same uncertainty method to EfficientNet, DenseNet, InceptionNet									X	X	X
6. Final Evaluation & Reporting	Aggregate results, prepare visualizations and documentation										X	X

18-NOV-2025

Summary:

- Subset ISIC Clinical Closeups
- Finetune ResNet50 on ISIC Clinical close-ups
- Run variational last layers on new finetuned model + New dataset

Blockers:

- Need to regain access to VMs on GCloud

Next Steps:

- Variational complete MLP
- Look into refining segmentations masks for better edge quality
- + XGBoost model based on extracted features (supplementary explainable model)
 - Maybe with PyRadiomics
- Aim for 99% Accuracy
- Waterloo images (should have classification)

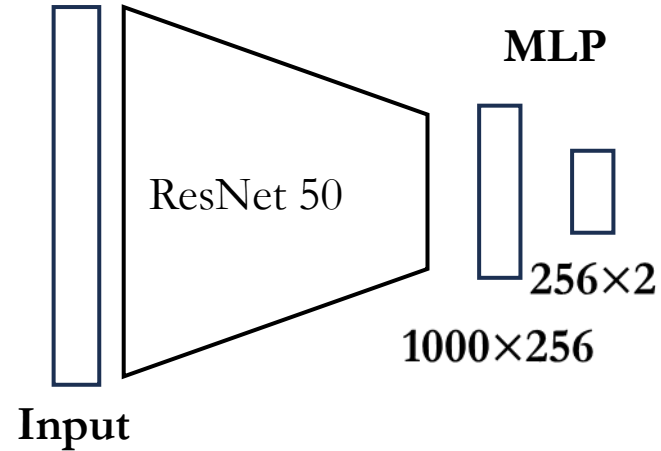
ISIC – Clinical Close up

- Total # of images: 9316
 - Benign: 3196
 - Malignant: 4787
 - Indeterminate: 1333
- Split data into
 - 70: 10: 20 (train: val: test)
 - Ensure that distribution of labels is even within each split



ResNet50

- Pretrained on ImageNet
- Finetune for 40 epochs
 - LR = $1e-4$
 - Batch size = 64
- Test accuracy: 87%
- Variational Last layer:
 - Test Accuracy: 87%



26-NOV-2025

Summary:

- Finetune ResNet50 on ISIC Clinical close-ups
- Run ResNet50 from random initialization on ISIC clinical close ups
- Run variational last layers on new finetuned model + New dataset
- Focusing on ensuring the model is learning proper probabilities – and now showing overconfidence in its **wrong** predictions
 - effect of pretrained model vs. trained from random init
 - deterministic vs. variational model

Blockers:

1/27/2026

MedirAI - Sprint Updates

- Need to regain access to VMs on Gcloud – waiting on pasindu for troubleshooting

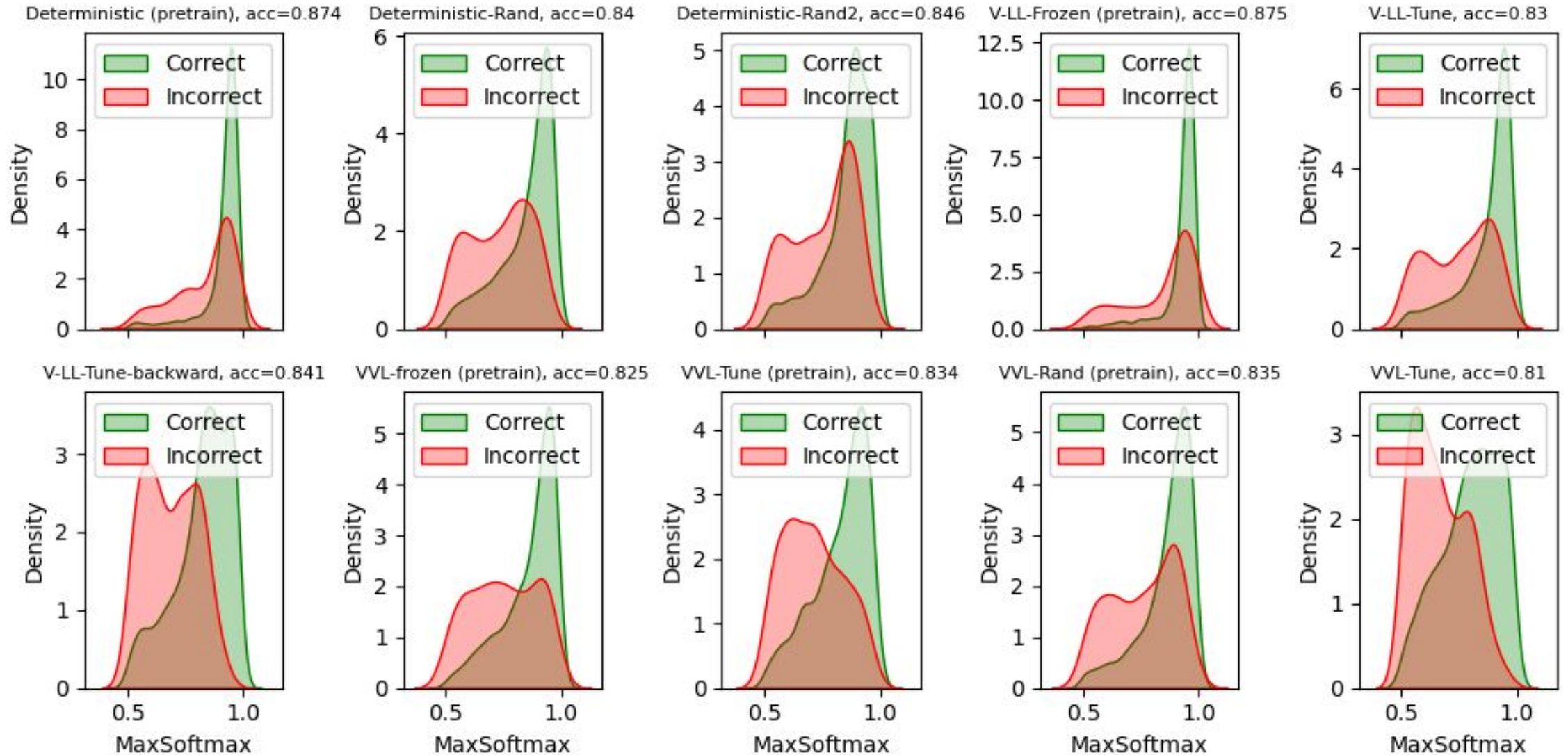
26-NOV-2025

Project Plan

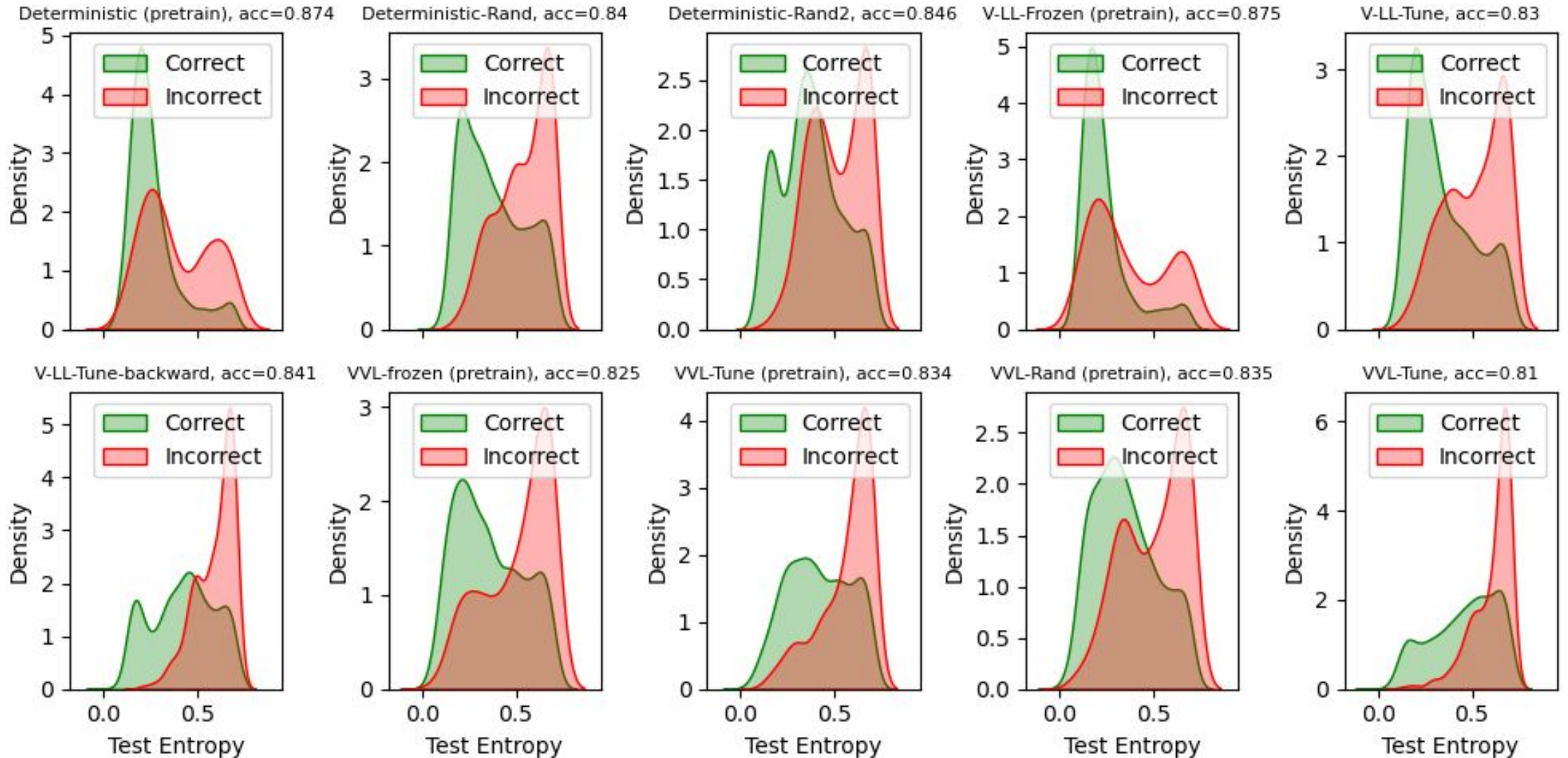
		October				November				December		
Task	Description	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3
1. Project Familiarization	Review current codebase, structure, data preprocessing, training pipeline	X	X	X	X							
2.A. Finetune ResNet50 (ISIC 2024 / Melanoma Kaggle)	Train and validate baseline deterministic ResNet50				X	X						
2.B. Finetune ResNet50 on ISIC Clinical close ups								X				
3. Variational Layers into ResNet	Add stochastic/variational layers (visible & hidden layers)					X	X	X				
4. Compare to Existing Methods (Grad-CAM)	Evaluate uncertainty visualization and compare with Grad-CAM methods								X	X		
5. Extend Uncertainty to Other Architectures	Apply same uncertainty method to EfficientNet, DenseNet, InceptionNet									X	X	X
6. Final Evaluation & Reporting	Aggregate results, prepare visualizations and documentation										X	X

26-NOV-2025 – maxSoftmax probabilities

effect of pretrained model vs. trained from random init
deterministic vs. variational model



26-NOV-2025 – test entropy



10-DEC-2025

Summary:

- Focusing on ensuring the model is learning proper probabilities – and now showing overconfidence in its **wrong** predictions
 - Ran variational first layer (as it has higher variability across samples for better detection)
 - Running ensemble of ResNet50 / ResNet18 for improving baseline performance
- Issues resolved regarding gcloud VM SSH through vscode

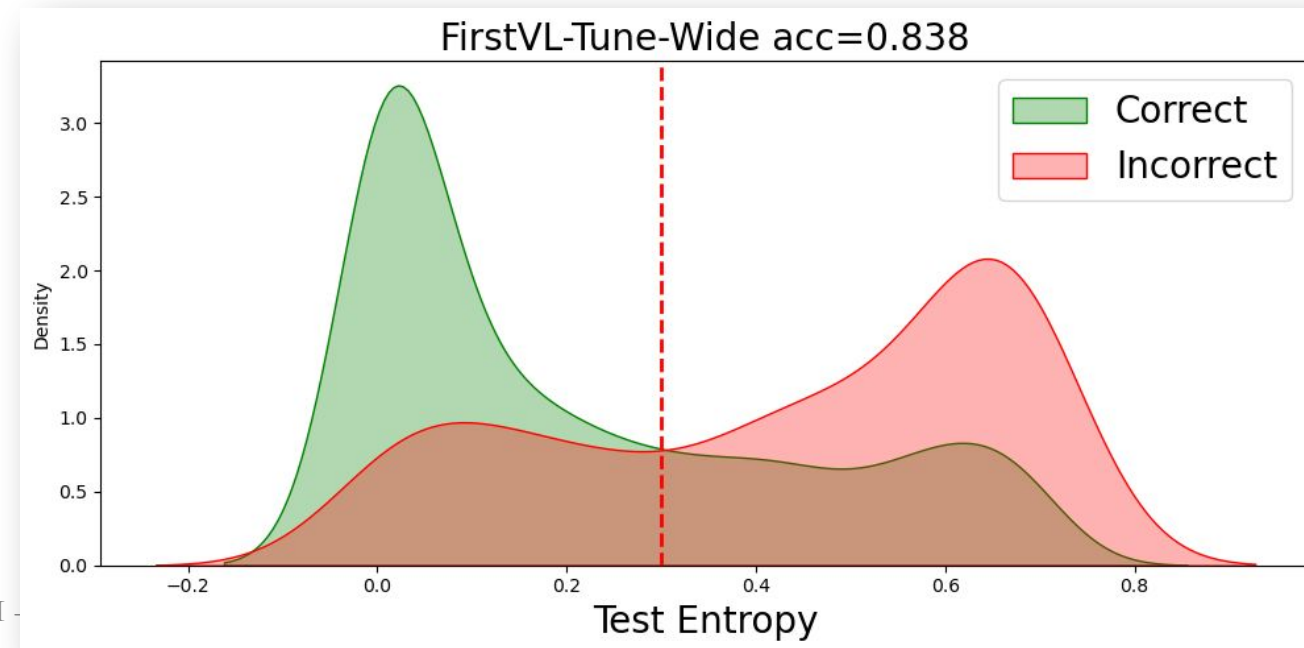
Blockers:

- None

1/27/2026
Nest Steps:

10-DEC-2025 – Variational First Layers

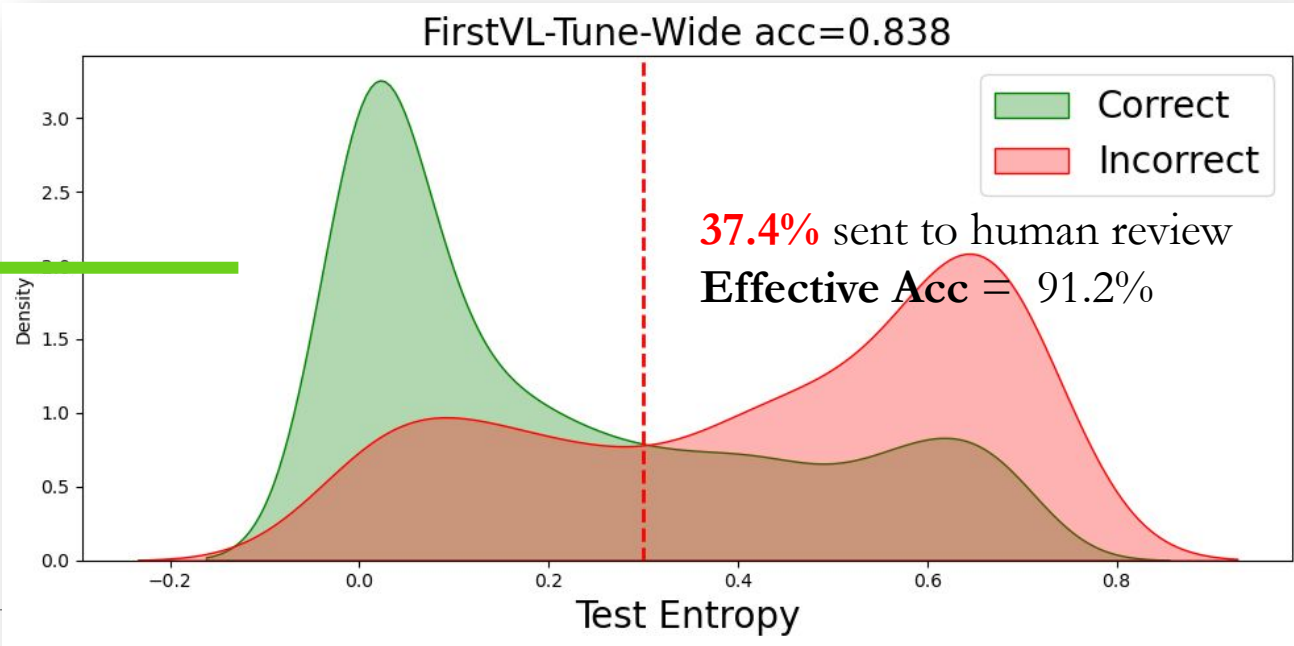
- Measure of Uncertainty: Entropy = $-p \log p$



10-DEC-2025 – Variational First Layers

- Measure of Uncertainty: Entropy of Expected = $mean\left(\sum_c(-p_{\mu_N} \log p_{\mu_N})\right)$

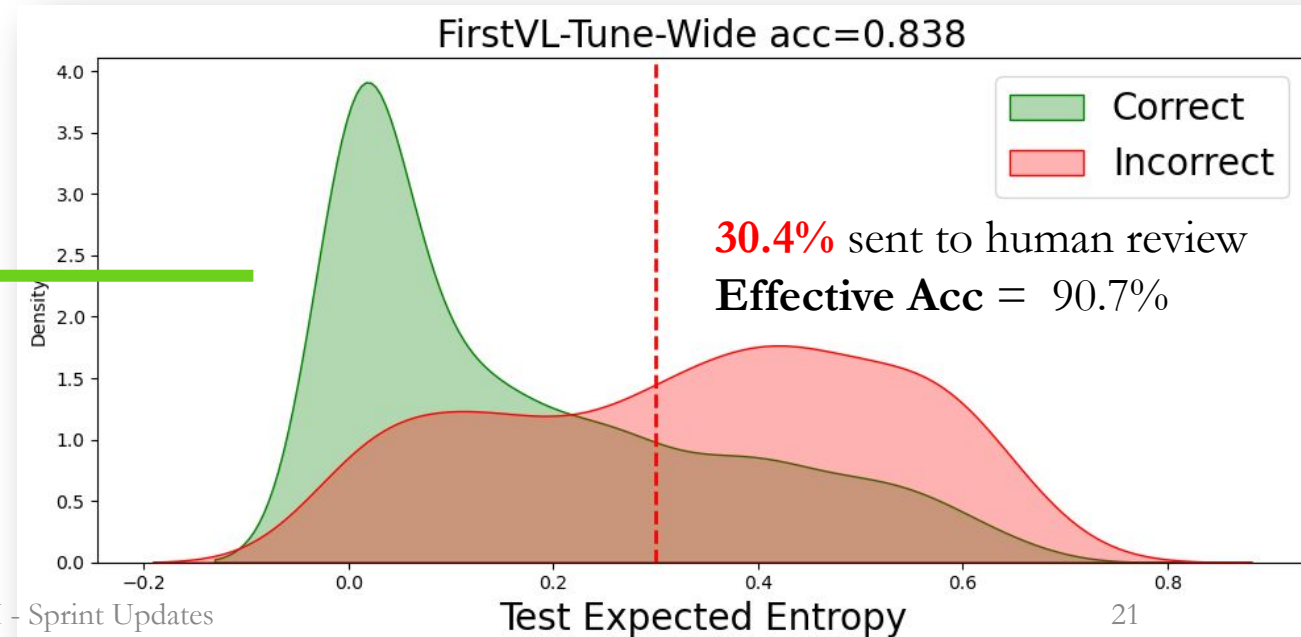
Entropy of Expected 1000/1597 597 (37.4% HR)		Predicted	
		Positive	Negative
Actual	Positive	515	37 (3.7%)
	Negative	4.5 (4.5%)	403



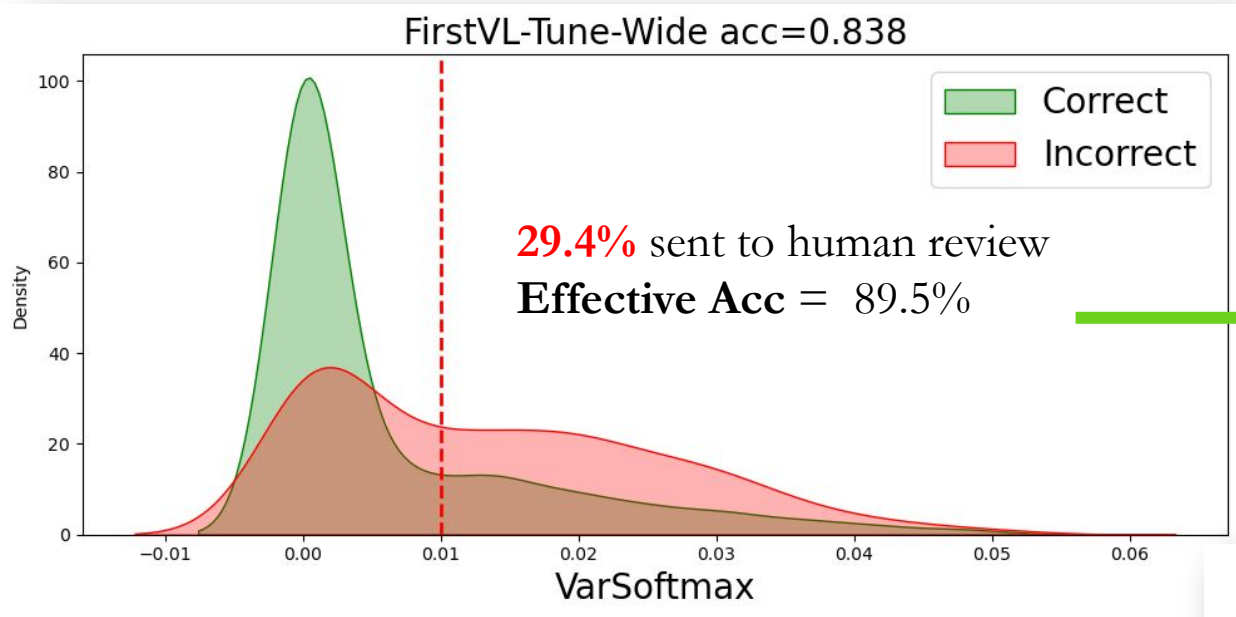
10-DEC-2025 – Variational First Layers

- Measure of Uncertainty: Expected Entropy = $mean(sum_c(-p \log p))$

Expected Entropy 1112/1597 485 (30.1% HR)		Predicted	
		Positive	Negative
Actual	Positive	577	48 (4%)
	Negative	55 (5%)	432

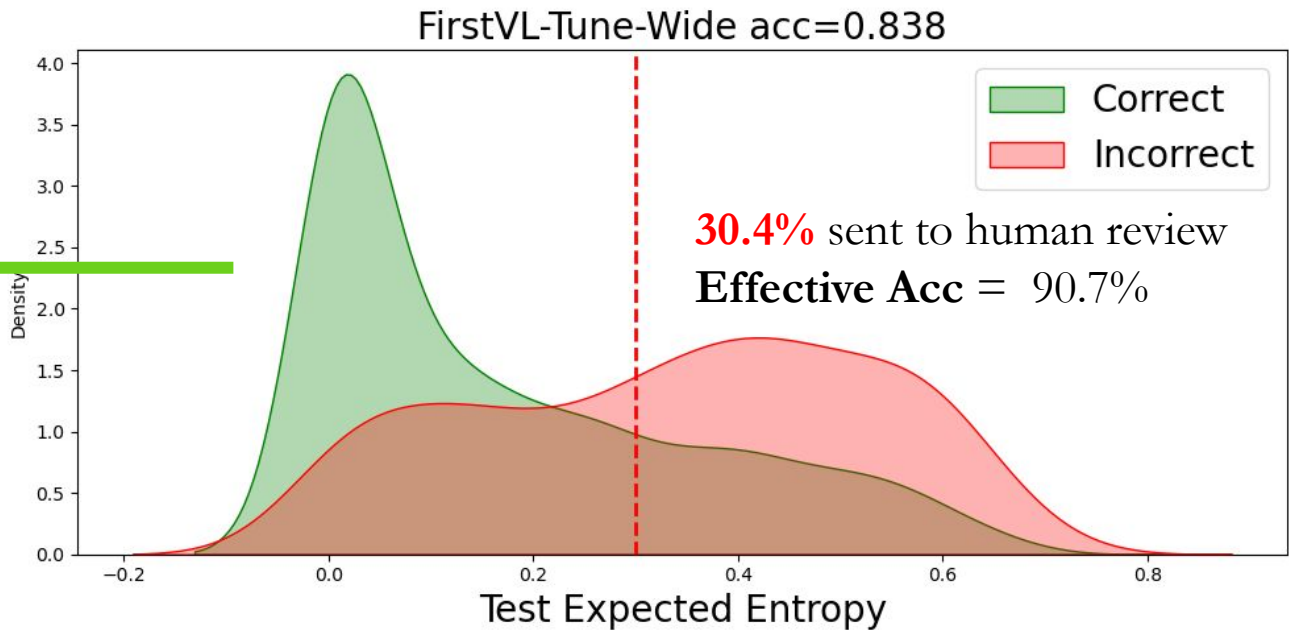


10-DEC-2025 – Variational First Layers



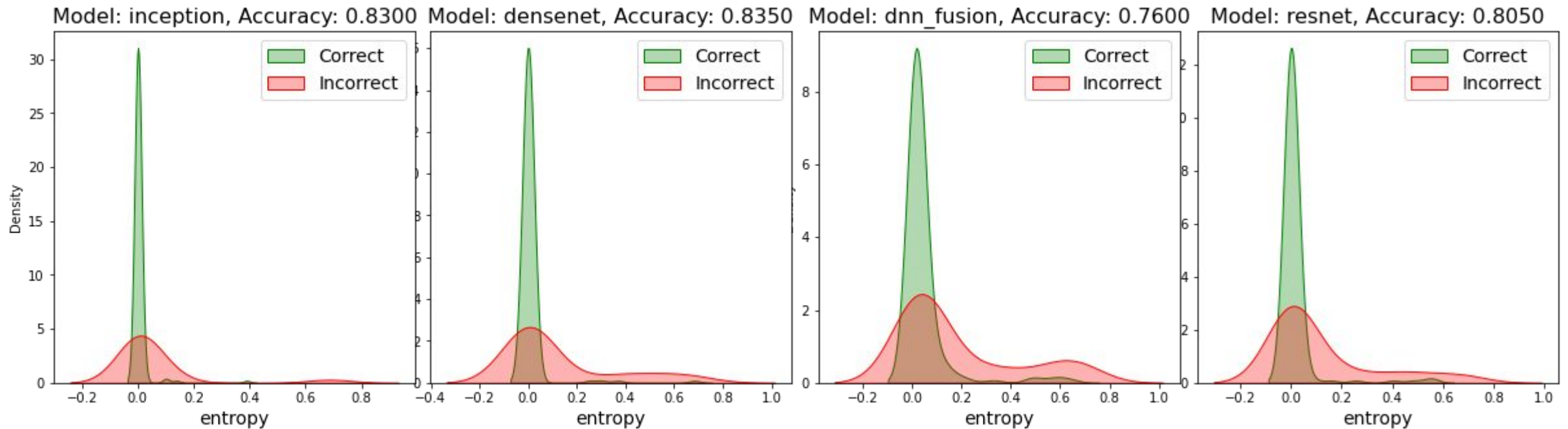
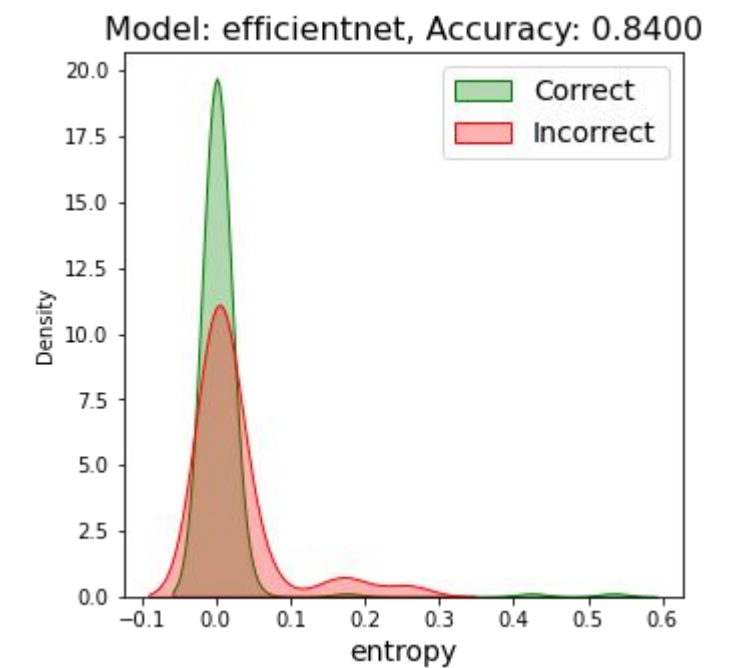
		Predicted	
		Positive	Negative
Actual	Positive	593	51 (4.5%)
	Negative	67 (5.9%)	416

		Predicted	
		Positive	Negative
Actual	Positive	577	48 (4%)
	Negative	55 (5%)	432



13-JAN-2026

- Analyze Tyson's models performance



13-JAN-2026

- Analyze Tyson's models performance

Model: clip, Accuracy: 0.5150
Model: derm, Accuracy: 0.5100
Model: densenet, Accuracy: 0.8350
Model: efficientnet, Accuracy: 0.8400
Model: inception, Accuracy: 0.8300
Model: resnet, Accuracy: 0.8050
Model: dnn_fusion, Accuracy: 0.7600
Model: found_fusion, Accuracy: 0.5050
Model: global_fusion, Accuracy: 0.5200

- Feature Fusion: Fine-Tuned
 - Accuracy: 0.905
 - False Negative Rate: 0.15
 - Confusion Matrix:

PanDerm

- Added MLP network to perform classification:
- Confusion Matrix:
 - Accuracy: 0.97
 - False Negative Rate: 0.06

Feature Fusion Quick Results

- Summary Statistics
 - Acc: 0.875
 - Confusion Matrix:

[100	0]	[TP
[25	75]	[FN

DermFoundation

- Added MLP network to perform classification:
- Confusion Matrix:
 - Accuracy: 0.955
 - False Negative Rate: 0.0 (will be higher in practice)
 - Confusion Matrix:

[91	9]	[TP	FP]
[0	100]	[FN	TN]

20-JAN-2026

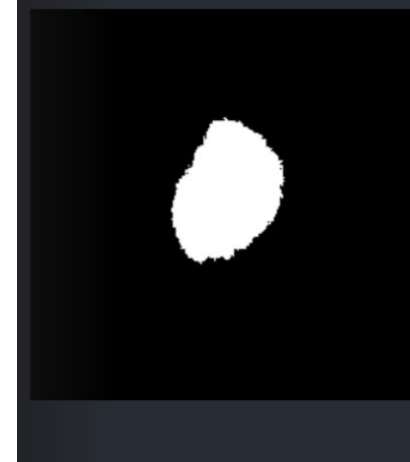
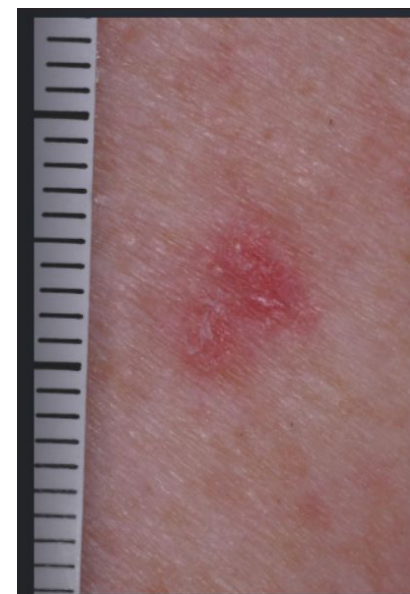
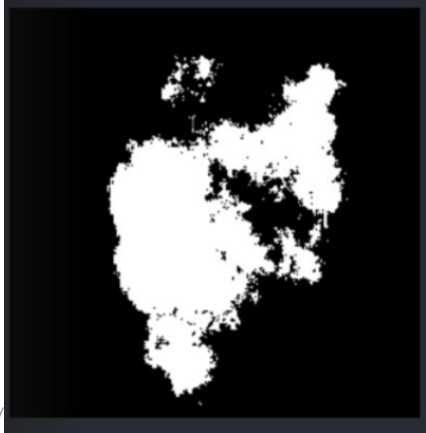
Summary:

- Reran experiments on Gcloud (after losing access to original servers where experiments were run)
- Running, Uncertainty Quantification with the Empirical Neural Tangent Kernel (NUQL)– NeurIPS25 – which shows high separability for failure detection
 - **Run with CIFAR-10 running successfully**
 - **ISIC run experiences issues**
- Filtering through pseudo segmentation masks for developing the decision tree classifier

Blockers:

- OOM error experienced with the ISIC run for NUQLs package (not experienced with baseline)

Pseudo Segmentation labels (by Aloys)



27-JAN-2026

Summary:

- Summarizing all experiments – (to be shared on google drive)
- Running segmentation uncertainty with Aloys's segmentation model (IP)
 - Uncertainty with UNet Model available
- New method for flagging for human review based on multiple model predictors.

Blockers:

- None

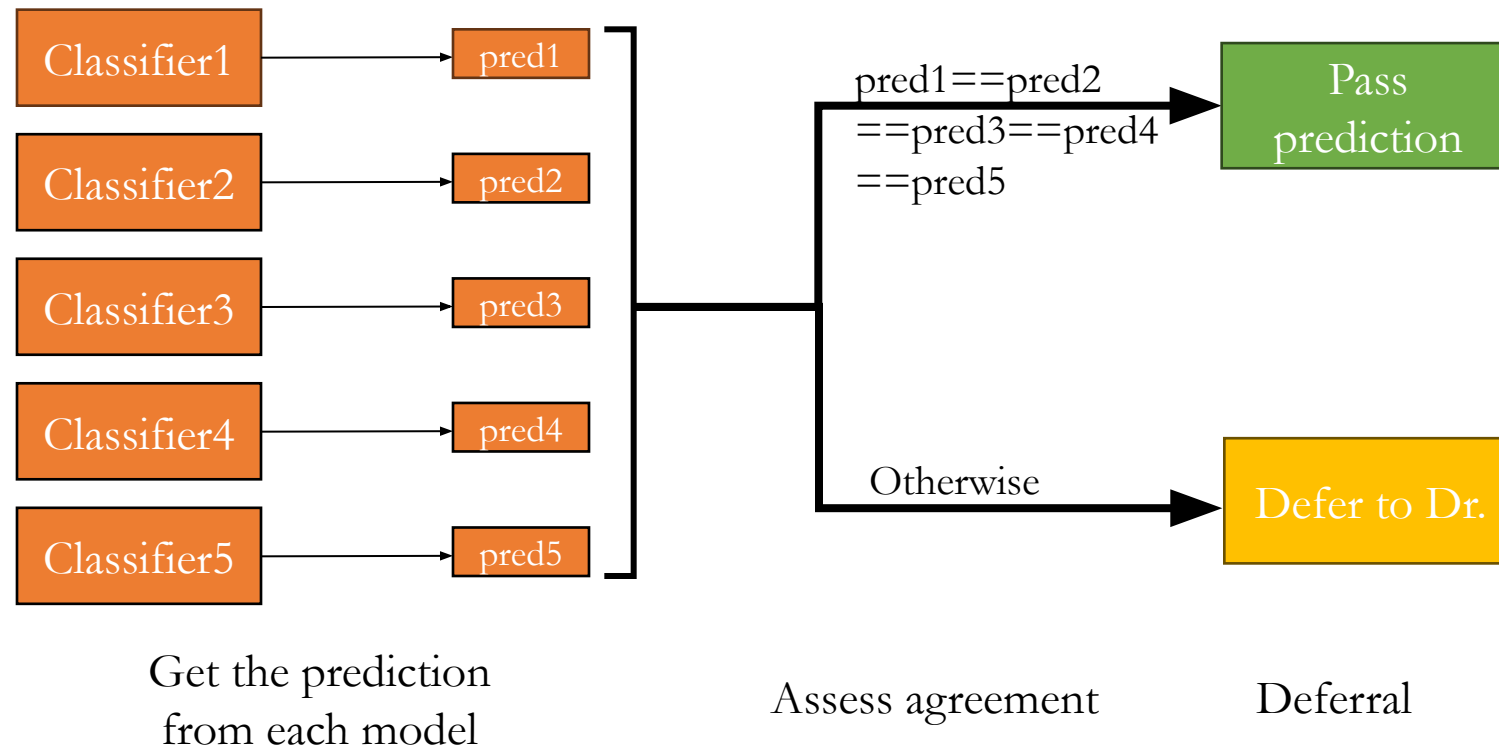
27-JAN-2026

- Combine predictions from multiple weak classifiers

Sample#	Classifier#1_pred	Classifier#2_pred	Classifier#3_pred	Classifier#4_pred	Prediction	SendToHuman Review
1	1	1	1	1	1	N/A
2	0	0	0	0	0	N/A
3	1	1	1	0	None	Yes
4	0	0	0	0	1	N/A
5	1	1	1	1	1	N/A

27-JAN-2026

Steps of deferral to human review:



27-JAN-2026

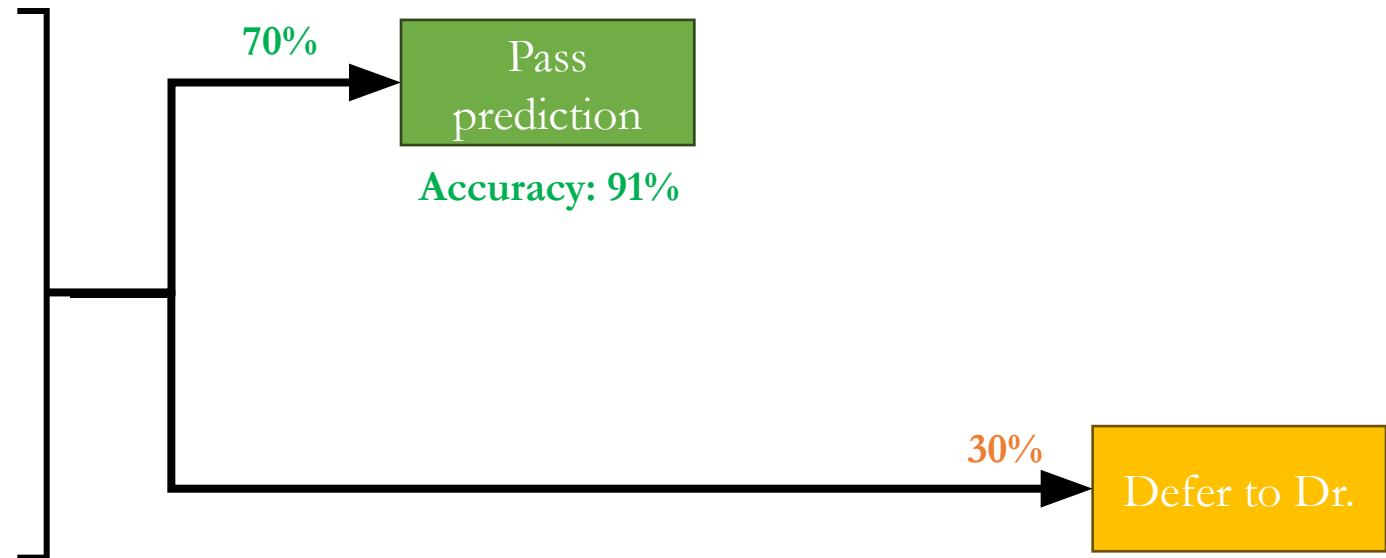
Steps of deferral to human review:

1) Get the prediction from each model

2) Assess agreement

3) Deferral

1) densenet, Accuracy: 0.84
2) efficientnet, Accuracy: 0.84
3) inception, Accuracy: 0.83
4) resnet, Accuracy: 0.81
5) dnn_fusion, Accuracy: 0.76
No model above 85%



27-JAN-2026

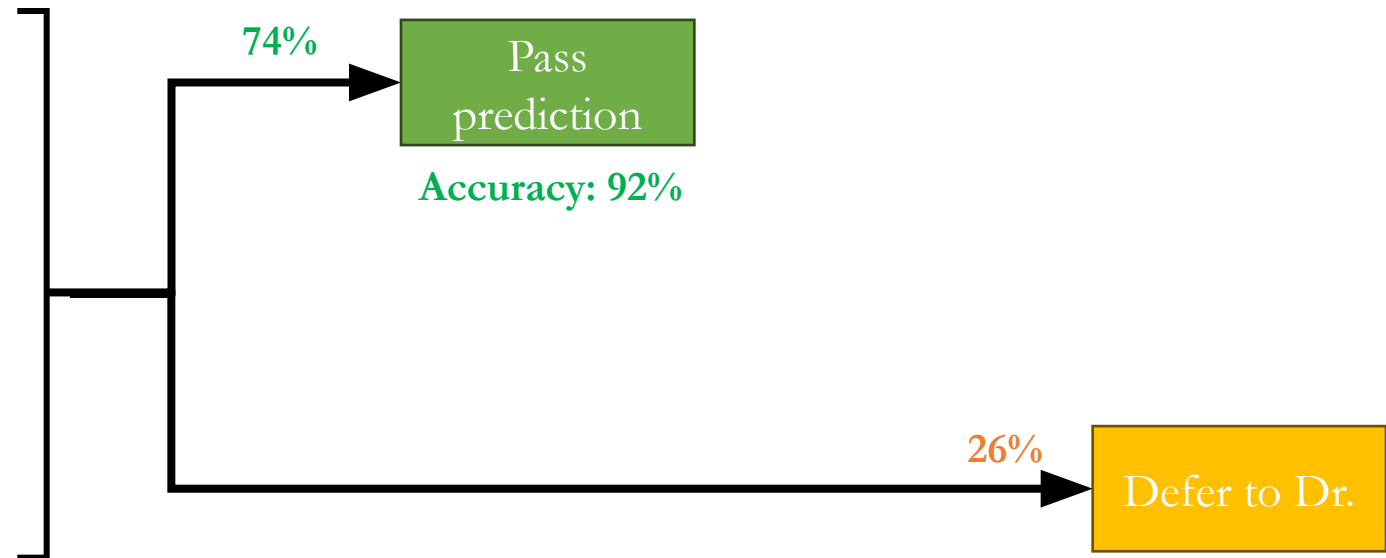
Steps of deferral to human review:

1) Get the prediction from each model

2) Assess agreement

3) Deferral

1) densenet, Accuracy: 0.84
2) efficientnet, Accuracy: 0.84
3) inception, Accuracy: 0.83
4) resnet, Accuracy: 0.81
~~5) dnn_fusion, Accuracy: 0.76~~
No model above 85%



27-JAN-2026

Why is this solution useful:

- 1. Threshold-free method:** Avoid thresholding based on varying entropies or variance of maxSoftmax, since this threshold could vary with distributional shifts.
- 2. Hard Constraint:** Controlled constraint on agreement of models: a hard constraint based on the models all agreeing
- 3. Signal for performance shift:** The deferral rate (increasing or decreasing) can be used to determine whether there's a shift in the dataset and that the underlying models' performance might be compromised.